Mehboob Alam

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Education

M.S. in Mathematics August 2025 – Expected May 2027
New Mexico State University, Las Cruces, NM, USA

- *GPA*: 3.50/4.00
- *Completed Coursework*: Advanced Linear Algebra, Abstract Algebra I, Introduction to Real Analysis I & II, and Statistics: Theory and Applications.
- *Coursework in Progress (Fall 2026)*: Machine Learning I, Probability: Theory and Applications, and Measure and Integration.

M.S. in Engineering Sciences (Applied Mathematics) February 2022 – January 2024
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan

- *GPA*: 3.75/4.00
- *Selected Coursework*: Analytical Solution of Partial Differential Equations for Engineers and Computational Methods for Engineers.
- *Thesis*: *Radial Basis Function Collocation Method for Fractional Partial Differential Equations*
- *Advisor*: Prof. Dr. Sirajul Haq

B.S. in Mathematics September 2016 – November 2020
University of Peshawar, Peshawar, Pakistan

- *GPA*: 3.72/4.00
- *Thesis*: *Stability Analysis of Impulsive Fractional and q-Fractional Implicit Differential Equations*
- *Advisor*: Prof. Dr. Akbar Zada

Research Interests

- Numerical Analysis and Scientific Computing
- Radial Basis Function and Meshless Methods
- Fractional Differential Equations
- Machine Learning
- Mathematical Modeling

Publications

Journal Articles

[J1] Zada, A., Riaz, U., Jamshed, J., **Alam, M.**, & Kallekh, A. (2025). Analysis of impulsive Caputo fractional integro-differential equations with delay. *Mathematical Methods in the Applied Sciences*, **48**(2), 2102–2121.

[J2] **Alam, M.**, Zada, A., & Abdeljawad, T. (2024). Stability analysis of an implicit fractional integro-differential equation via integral boundary conditions. *Alexandria Engineering Journal*, **87**, 501–514.

[J3] **Alam, M.**, Haq, S., Ali, I., Ebadi, M. J., & Salahshour, S. (2023). Radial Basis Functions Approximation Method for Time Fractional FitzHugh-Nagumo Equation. *Fractal and Fractional*, **7**(12), 882.

[J4] Amin, F., Waqas, H. A., **Alam, M.**, Yousafzai, A., Naeem, K., Khan, N., Gerges, M., & Fawad, M. (2023). Strengthening of Reinforced Concrete Deep Beams with Large Openings Using Metal Plates. *Emirati Journal of Civil Engineering and Applications*, **1**(1), 31–44.

[J5] **Alam, M.**, Zada, A., Begum, S., & Riaz, U. (2023). Analysis of Fractional Integro-differential System with Impulses. *International Journal of Applied and Computational Mathematics*, **9**, 93.

[J6] Majeed, R., Zhang, B., & **Alam, M.** (2023). Fractional Langevin Coupled System with Stieltjes Integral Conditions. *Mathematics*, **11**, 2278.

[J7] **Alam, M.**, & Khalid, K. H. (2023). Analysis of q-fractional coupled implicit systems involving the nonlocal Riemann–Liouville and Erdélyi-Kober q-fractional integral conditions. *Mathematical Methods in the Applied Sciences*, **46**(12), 12711–12734.

[J8] **Alam, M.**, Khan, A., & Asif, M. (2023). Analysis of implicit system of fractional order via generalized boundary conditions. *Mathematical Methods in the Applied Sciences*, **46**(9), 10554–10571.

[J9] Guo, L., Riaz, U., Zada, A., & **Alam, M.** (2023). On Implicit Coupled Hadamard Fractional Differential Equations with Generalized Hadamard Fractional Integro-Differential Boundary Conditions. *Fractal and Fractional*, **7**(1), 13.

[J10] Zhang, B., Majeed, R., & **Alam, M.** (2022). On Fractional Langevin Equations with Stieltjes Integral Conditions. *Mathematics*, **10**, 3877.

[J11] Zada, A., **Alam, M.**, Khalid, K. H., Iqbal, R., & Popa, I. L. (2022). Analysis of Q-Fractional Implicit Differential Equation with Nonlocal Riemann–Liouville and Erdélyi-Kober Q-Fractional Integral Conditions. *Qualitative Theory of Dynamical Systems*, **21**, 93.

[J12] **Alam, M.**, & Zada, A. (2022). Implementation of q-calculus on q-integro-differential equation involving anti-periodic boundary conditions with three criteria. *Chaos, Solitons & Fractals*, **154**, 111625.

[J13] **Alam, M.**, Zada, A., & Riaz, U. (2022). On a Coupled Impulsive Fractional Integrodifferential System with Hadamard Derivatives. *Qualitative Theory of Dynamical Systems*, **21**, 8.

[J14] **Alam, M.**, Zada, A., Popa, I. L., Kheiryan, A., Rezapour, S., & Kaabar, M. K. A. (2021). A Fractional Differential Equation with Multi-Point Strip Boundary Condition involving the Caputo Fractional Derivative and its Hyers–Ulam Stability. *Boundary Value Problems*, **2021**, 73.

[J15] **Alam, M.**, & Shah, D. (2021). Hyers–Ulam stability of coupled implicit fractional integro-differential equations with Riemann–Liouville derivatives. *Chaos, Solitons & Fractals*, **150**, 111122.

[J16] Luo, D., **Alam, M.**, Zada, A., Riaz, U., & Luo, Z. (2021). Existence and stability of implicit fractional differential equations with Stieltjes boundary conditions having Hadamard derivatives. *Complexity*, **2021**(3), 1–36.

[J17] Wang, X., **Alam, M.**, & Zada, A. (2021). On coupled impulsive fractional integro-differential equations with Riemann–Liouville derivatives. *AIMS Mathematics*, **6**(2), 1561–1595.

[J18] Zada, A., **Alam, M.**, & Riaz, U. (2021). Analysis of q-fractional implicit boundary value problems having Stieltjes integral conditions. *Mathematical Methods in the Applied Sciences*, **44**(6), 4381–4413.

Submitted Manuscripts

[S1] RBFs approximation method for time fractional Rosenau–Hyman equation.

[S2] Analysis of nonlocal implicit system of fractional order via Riemann–Liouville integral conditions.

Research Projects

Radial Basis Function Collocation Methods for Fractional PDEs

- Implemented meshfree radial basis function (RBF) collocation methods in MATLAB for the numerical solution of nonlinear time-fractional partial differential equations.
- Combined RBF spatial discretization with a θ-weighted finite difference scheme, using a Caputo fractional derivative approximation of order $\mathcal{O}(\Delta t^{2-\alpha})$, $0 < \alpha < 1$.
- Applied the numerical schemes to Rosenau–Hyman, FitzHugh–Nagumo, and generalized Burgers–Huxley and Burgers–Fisher equations, including one- and two-dimensional spatial problems.
- Investigated multiquadric (MQ), inverse multiquadric (IMQ), inverse quadratic (IQ), and Gaussian RBFs, employing an extended Rippa algorithm for shape-parameter selection.
- Examined numerical stability through amplification-matrix eigenvalue analysis and evaluated accuracy using L_2, L_∞, and L_{rms} error norms, with comparisons against exact solutions and existing numerical methods.
- Published thesis-related findings on the time-fractional FitzHugh–Nagumo equation in *Fractal and Fractional* (2023) [J3].

Finite Element Analysis of Reinforced Concrete Deep Beams

- Applied finite element modeling in Abaqus to investigate the structural behavior and strengthening of reinforced concrete deep beams with large openings using metal plates.
- Evaluated model validation and mesh sensitivity against experimental results, obtaining a numerical peak load of 430 kN compared with an experimental value of 410 kN.
- Investigated four metal-plate configurations under a 500 kN load, analyzing their effects on structural deflection and load-carrying capacity.
- Identified the configuration with the lowest deflection (approximately 0.45 mm) and assessed the cost-effectiveness of alternative plate arrangements and concrete grades.
- Coauthored the corresponding research article published in *Emirati Journal of Civil Engineering and Applications* (2023) [J4].

Optimization of Resource Allocation in Railway Networks

- Investigated the operational efficiency and staff allocation of 44 railway stations in Khyber Pakhtunkhwa, Pakistan, using Data Envelopment Analysis (DEA).
- Formulated a hybrid approach combining Parabolic Data Envelopment Analysis (PDEA) with a Sequential Algorithm (SA) to obtain integer-valued staff allocations while preserving the total workforce.
- Analyzed resource redistribution among 858 employees using an input-oriented BCC efficiency model and a two-stage optimization procedure.
- Improved modeled average system efficiency from 56% to 99.5% through optimized staff redistribution across the railway stations.

Existence, Uniqueness, and Stability of Fractional and q-Fractional Differential Equations

- Investigated the existence, uniqueness, and stability of solutions to nonlinear fractional and q-fractional differential equations, including coupled impulsive systems and nonlocal boundary value problems.
- Established sufficient conditions for existence and uniqueness using Banach's contraction principle and fixed-point theorems of Krasnoselskii, Schaefer, and Leray–Schauder.
- Derived Hyers–Ulam, generalized Hyers–Ulam, Hyers–Ulam–Rassias, and generalized Hyers–Ulam–Rassias stability results using nonlinear functional analysis.
- Analyzed systems involving Caputo and Hadamard fractional derivatives and Stieltjes integral boundary conditions, illustrating theoretical results through mathematical examples.

Machine Learning Course Projects

Scientific Programming and Data Analysis with Julia

- Developed modular Julia programs for scientific computing, incorporating numerical algorithms, matrix operations, data structures, and reusable functions.

- Processed and visualized real-world datasets, including 272 Old Faithful geyser observations and 200 neural spike recordings, to explore data distributions and patterns through scientific visualization.
- Implemented computational algorithms for minimum-coin change optimization and geometric point classification using vector operations and cross-product techniques.
- Organized computational workflows using Julia modules and packages, generating reproducible visualizations and technical reports with Plots.jl and Weave.jl.

Unsupervised Learning and K-Means Clustering

- Implemented batch and online K-means clustering algorithms from scratch in Julia, incorporating distance-based cluster assignment, centroid updates, and convergence criteria.
- Developed histogram-based density estimation and multivariate Gaussian sampling algorithms to generate, visualize, and analyze synthetic datasets.
- Conducted Monte Carlo experiments with up to 500 trials to evaluate clustering accuracy, convergence behavior, centroid initialization, and objective-function performance; investigated learning-rate effects in online K-means.
- Applied K-means clustering to RGB image segmentation and color quantization using $K = 4$, 8, and 16 clusters, analyzing image quality, color representation, and the transferability of learned centroids across images.

Linear and Polynomial Regression with Regularization

- Implemented bivariate, multivariate, and polynomial least-squares regression algorithms from scratch in Julia, using analytical coefficient estimation and matrix pseudoinverse methods.
- Developed synthetic linear and nonlinear datasets with Gaussian noise to investigate parameter estimation and visualize fitted regression models in two and three dimensions.
- Evaluated polynomial models of degrees $M = 0$ to 9 using training and testing root-mean-square errors (RMSE), investigating model complexity, training sample size, and overfitting.
- Implemented L_2-regularized polynomial regression and analyzed the effects of regularization strength on coefficient magnitudes, model flexibility, and generalization behavior.

Supervised Classification and Decision Boundary Analysis

- Implemented K-Nearest Neighbors (KNN) and multiclass least-squares classification algorithms from scratch in Julia, incorporating distance-based majority voting, one-hot encoding, and matrix pseudoinverse methods.
- Developed synthetic classification datasets, including Double Moon, Concentric Gaussian, and Gaussian XOR distributions, to investigate nonlinear class structures and visualize decision boundaries.
- Evaluated KNN classifiers for $K = 1$, 5, and 15 using training sets of $N = 50$ and 500 samples, analyzing the effects of neighborhood size and data availability on classification boundaries.
- Achieved training accuracies of 97.2% on Gaussian data, 100% on Dataset A, 96.25% on Dataset A2, and 99% on Dataset B using least-squares classification, evaluating binary and multiclass decision regions.

Professional Qualifications

Primary Teaching Certificate 2016

Allama Iqbal Open University, Islamabad, Pakistan

- *Professional Training*: Constructivist teaching methodologies, classroom management, student assessment, educational technology integration, and curriculum development.

Professional Service

Peer Review Experience

- Completed 25 verified manuscript reviews across 15 scholarly journals (2021–2026).
- *Web of Science Researcher Profile*: PCT-8163-2025

Journals Reviewed

- Abstract and Applied Analysis
- Beni-Suef University Journal of Basic and Applied Sciences
- Boundary Value Problems
- Communications in Nonlinear Science and Numerical Simulation
- Chaos, Solitons & Fractals
- Complex Analysis and Operator Theory
- Computational Mathematics and Modeling
- International Journal of Applied and Computational Mathematics
- Journal of Applied Mathematics
- Journal of Applied Mathematics and Computing
- Journal of Mathematical Sciences
- Mathematical Problems in Engineering
- Nonlinear Analysis: Modelling and Control
- Scientific Reports
- Vietnam Journal of Mathematics

Teaching Experience

Graduate Teaching Assistant August 2025 – Present

New Mexico State University, Las Cruces, NM, USA

- *MATH 3460:* Introduction to Ordinary Differential Equations Spring 2026, Fall 2026
- *MATH 2415:* Introduction to Linear Algebra Fall 2025
- Graded student submissions and provided feedback for the departmental Math Problem of the Week program Fall 2026

Lecturer February 2024 – March 2025
The University of Lahore, Sargodha, Pakistan

- *MATH-402:* Partial Differential Equations — Spring 2024
- *MAT07301:* Complex Analysis — Spring 2024
- *MAT12145:* Introduction to Mathematical Modeling — Spring 2024, Fall 2024
- *MAT09212/CS09202:* Discrete Mathematics / Discrete Structures — Spring 2024, Fall 2024
- *MATH2100/MA2334:* Ordinary Differential Equations — Summer 2024
- *MATH-204:* Metric Space and Group Theory — Summer 2024
- Infinite Series and Sequences — Summer 2024
- Probability and Statistics — Summer 2024
- Multivariate Calculus — Fall 2024
- *Administrative Responsibility:* Served as Timetable Coordinator, managing academic scheduling and supporting course coordination.

Graduate Teaching Assistant February 2022 – January 2024
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan

- *ES202:* Engineering Statistics — Spring 2022
- *MT203:* Complex Variables and Transforms — Spring 2022, Spring 2023
- *MT102:* Differential Equations and Linear Algebra I — Summer 2022
- *MT101:* Calculus I — Fall 2022
- *ES341:* Numerical Analysis — Fall 2022, Fall 2023

Visiting Lecturer November 2021 – February 2022
Government Postgraduate College Nowshera, Pakistan

- Basic Algebra — Fall 2021
- Differential Geometry — Fall 2021
- Partial Differential Equations — Fall 2021

Visiting Lecturer March 2021 – August 2021
Government Degree College Pabbi, Nowshera, Pakistan

- Basic Algebra — Spring 2021
- Partial Differential Equations — Spring 2021
- Measure Theory — Spring 2021
- Classical Mechanics — Spring 2021

Research Supervision

Graduate Research Supervision Fall 2024 – Spring 2025
The University of Lahore, Sargodha, Pakistan

- **Ammara Ijaz** (M.Phil. Mathematics)
  Thesis: Qualitative Analysis of Fractional Langevin Equation.

Undergraduate Research Co-Supervision Spring 2021
Government Degree College Pabbi, Nowshera, Pakistan

- **Group 1:** Muhammad Irfan and Farid Ullah.
  Research Topic: Existence and Stability Analysis of Three Point Boundary Value Problem.
- **Group 2:** Muhammad Asif, Ali Abbas, and Atif.
  Research Topic: Analysis of Implicit System of Fractional Order via Generalized Boundary Conditions.
  Research Outcome: Coauthored publication with Muhammad Asif [J8] (2023).
- **Group 3:** Akhtar Azam, Muhammad Abbas, and Izaz Ali.
  Research Topic: Stability Analysis of Fractional Integro-Differential Boundary Value Problem.
  Research Outcome: Related manuscript by Akhtar Azam, submitted for publication [S2].

Presentations

Graduate Course Presentations
Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Topi, Pakistan

- *Existence and Uniqueness of Solutions to Differential Equations*.
  Course: Analytical Solution of Partial Differential Equations for Engineers — 17 December 2022.
- *A Simple Non-Chaotic Map Generating Subdiffusive, Diffusive, and Superdiffusive Dynamics*.
  Course: Special Topics in Applied Mathematics — 6 December 2022.
- *Analysis of Parabolic DEA and Sequential Algorithm: A Case Study in KPK Railway Stations*.
  Course: Operation Management — 2 June 2022.
- *Understanding the Performance Measurement of the Indian Steel Industry: A DEA Approach*.
  Course: Operation Management — 6 April 2022.

- *Improving Inland Freight Logistics Efficiency: Is There an Ideal Modal Split?*
  Course: Operation Management 9 March 2022.

Conference Attendance

- *13th Conference on Dynamical Systems Applied to Biology and Natural Sciences* 8–11 February 2022
  organized by the Mathematical and Theoretical Biology Group, Basque Center for Applied Mathematics, Bilbao, Spain (Virtual Attendance).
- *International Conference on Recent Trends in Applied Mathematics* 18 May 2022
  organized by the Department of Mathematics, University of Engineering and Technology, Lahore, Pakistan.

Awards & Distinctions

- **Certificate of Merit – Third Position in B.S. Mathematics** 2020
  University of Peshawar, Peshawar, Pakistan
- **HEC Laptop Award** 2019
  Higher Education Commission of Pakistan

Scholarships & Funding

- **Graduate Teaching Assistantship** Fall 2025 – Spring 2027
  New Mexico State University, Las Cruces, NM, USA
- **Graduate Assistantship** 2022 – 2024
  GIK Institute of Engineering Sciences and Technology, Topi, Pakistan
- **Ehsaas Scholarship** 2019 – 2020
  University of Peshawar, Peshawar, Pakistan

Technical Skills

Programming & Scientific Computing

- Python, Julia, MATLAB, Mathematica

Machine Learning & Data Analysis

- Supervised and Unsupervised Learning, Regression, Classification, K-Means Clustering, K-Nearest Neighbors, Regularization, and Data Visualization

Numerical Methods & Computational Modeling

- Radial Basis Function Methods, Meshless Collocation, Finite Difference Methods, Numerical Stability Analysis, Finite Element Analysis, and Mathematical Optimization

Scientific Software & Documentation

- Abaqus, DEAFrontier, LaTeX, Microsoft Office, and Weave.jl

Languages

- **Pashto:** Native
- **Urdu:** Fluent
- **English:** Professional Working Proficiency

Leadership & Volunteer Activities

Finance Secretary August 2025 – Present
Pakistani Students Association (PSA), New Mexico State University

- Managed budget planning, expense tracking, and financial records for student-organized events and activities.
- Coordinated with the executive committee to support transparent financial reporting and responsible resource allocation.
- Supported cultural, academic, and community engagement initiatives promoting student involvement.

Volunteer Judge February 2026
Data Science & Analytics Competition, NM TSA State Conference

- Evaluated student projects in data science and analytics, assessing analytical approaches, problem-solving methods, and technical presentations.
- Provided constructive feedback to participants on their analytical work and presentation quality.

STEM Outreach Volunteer – Learn-A-Palooza November 2025
New Mexico State University, Las Cruces, NM, USA

- Supported the NMSU STEM Outreach Center in organizing an educational event, assisting with participant engagement and event logistics.
- Completed three hours of certified community service.